

Quiz (Carolina Reaper)**Q1.** What is the point-slope form of a linear equation?

- (A) $y - y_1 = m(x - x_1)$
 (B) $y = mx + b$
 (C) $ax + by = c$
 (D) $x = my + b$
 (E) $y = \frac{1}{m}x + b$

(A) $(-2, 0)$ (B) $(-1, 0)$

(C) 0

(D) 2

(E) $(0, -2)$ **Q2.** Which form is $2x + 3y = 5$?

- (A) Point-Slope Form
 (B) Slope-Intercept Form
 (C) Standard Form
 (D) Vertex Form
 (E) Parametric Form

Q6. What is the equation of a line with slope -2 and passes through $(1, 3)$?(A) $y = -2x + 1$ (B) $y = -2x + 5$ (C) $y = 2x - 1$ (D) $y = 2x + 1$ (E) $y = -2x + 3$ **Q3.** What is the slope of $y = 2x - 3$?

- (A) 2
 (B) -2
 (C) $\frac{1}{2}$
 (D) $-\frac{1}{2}$
 (E) 3

Q7. Convert $x + 2y = 4$ to slope-intercept form.(A) $y = \frac{1}{2}x + 2$ (B) $y = -\frac{1}{2}x + 2$ (C) $y = \frac{1}{2}x - 2$ (D) $y = -2x + 1$ (E) $y = \frac{1}{2}x - 1$ **Q4.** Convert $y = 2x - 3$ to standard form.

- (A) $2x - y = 3$
 (B) $2x + y = 3$
 (C) $-2x + y = 3$
 (D) $2x - y = -3$
 (E) $x + 2y = 6$

Q8. What is the equation of a line in point-slope form with slope 3 and passes through $(-1, 2)$?(A) $y - 2 = 3(x + 1)$ (B) $y + 2 = 3(x - 1)$ (C) $y - 2 = -3(x + 1)$ (D) $y + 1 = 3(x - 2)$ (E) $y - 1 = 3(x + 2)$ **Q5.** Graph $y = -x + 2$. What is the y-intercept?**Open Ended Questions (FRQ)****Q9.** Graph $y = 2x - 1$ and $x + 2y = 3$ on the same coordinate plane. Find the point of intersection.**Q10.** A water tank can hold 120 gallons of water. Due to a leak, the tank is being filled at a rate of 2 gallons per minute and is leaking at a rate of 1 gallon per minute. Write an equation to represent the amount of water in the tank after t minutes. Find the time it takes to fill the tank.**Chef's Tips**

- Tip 1: Use graph paper to help with the free response questions.
- Tip 2: Make sure to label all axes and lines clearly.
- Tip 3: Use the given equations to find the point of intersection.